

Read the article "Making Mona Lisa Smile." Then answer the question.

Making Mona Lisa Smile

by Nick D'Alto



Imagine being able to detect exactly how other people are feeling, even when they are trying to hide their emotions. New software has been developed that improves accuracy in perceiving the emotions of others. This new software, the Facial Action Coding System, has promising applications for a variety of industries.

❶ She's 83 percent happy, 9 percent disgusted, 6 percent fearful, and 2 percent angry. No, that's not your science teacher, grading your latest lab assignment. It's the subject of Leonardo da Vinci's Renaissance painting, *Mona Lisa*—at least according to some new computer software that can recognize emotions. The software is the latest innovation from Prof. Thomas Huang, of the Beckman Institute for Advanced Science at the University of Illinois, working in collaboration with Prof. Nicu Sebe of the University of Amsterdam. Dr. Huang and his colleague are experts at developing better ways for humans and computers to communicate.

❷ Hold on! Can we actually "calculate" emotions—like math homework? And how can a computer recognize the subtle facial movements we humans use to express how we feel?

❸ The process begins when the computer constructs a 3-D computer model of the face; all 44 major muscles in the model must move like human muscles. Movement of one or more muscles is called an "action unit." Then Dr. Huang relies on the work of psychologists, such as Dr. Paul Eckman, creator of FACS (Facial Action Coding System). Eckman has classified six basic emotions—happiness, surprise, anger, disgust, fear, and sadness—and then associated each with characteristic movements of the facial muscles. For example, your *frontalis pars lateralis* muscle (above your eyes) raises your eyebrows when you're surprised; your *orbicularis oris* (around your mouth) tightens your lips to show anger.

❹ "The facial expressions for each emotion are universal," observes Dr. Huang, "even though individuals often show varying degrees of expression" (like not smiling as broadly). Using video imagery, the new emotion-recognition software tracks these facial movements—in a real face or in the painted face of Mona Lisa. By weighting the different units, the software can even identify mixed emotions (as in da Vinci's masterpiece). Each expression is compared against a neutral face (showing no emotion).

❺ In fact, we humans perform this same impressive "calculation" every day. For instance, you can probably tell how a friend is feeling simply by the look on her face. Of course, most of us would have trouble actually describing each facial trait that conveys happy, worried, etc. Yet Dr. Huang observes that artists such as da Vinci studied human anatomy to help them paint facial muscles precisely enough to convey specific emotions. His new computer software stores similar anatomical information as electronic code. Perhaps Dr. Huang's emotion algorithms are a different sort of "Da Vinci Code"!

❻ The Mona Lisa demonstration is really intended to bring a smile to your face, while it shows just how much this computer can do. Imagine a computer that knows when you're happy or sad. For example, if you smile when a Web ad appears on your screen, a similar ad might follow. But if you frown, the next ad will be different. "A classroom computer could recognize when a student is becoming confused or bored," Dr. Huang predicts. "Then it could modify the lesson, like an effective human instructor." The same technology can make computer-animated faces more expressive—for video games or video surgery. "Most human communication is nonverbal, including emotional communication," notes Dr. Huang. "So computers need to understand that, too."

❼ Your home PC can't handle the complex algorithms used to decode Mona Lisa's smile. But we can write down some simple instructions that "encode" different emotions.

While looking in the mirror:

1. Raise your lips at the corners of your mouth.
2. Then squint your eyes slightly, to produce wrinkling ("crow's-feet") at the corners of your eyes.
3. Holding that, raise the outer parts of your cheeks up, toward your eyes.

Does your expression in the mirror suggest an emotion? Can your lab partner recognize which one? In fact, these are the instructions for a face that looks happy. It's all about those muscular action units. They even indicate the difference between a genuine smile and a forced one. In the real smile, the *zygomatic major* (muscles that begin at your cheek bones) lift the corners of your mouth. (That's our instruction #1.)

❶ Meanwhile, muscles called *orbicularis oculi pars palpebrae* make crow's-feet around your eyes. (That's instruction #2.) But in a false smile, the mouth is stretched sideways using the *zygomatic major* and a different muscle, the *risorius*. To an expert, faces don't lie; these muscle clues are sometimes used to spot when a "smiling" politician or celebrity isn't being truthful.

❷ By the way, did making a happy face in this experiment also make you *feel* slightly happy? According to the Facial Feedback Theory of Emotion, moving your facial muscles not only expresses emotions, but also may even help produce them. ("Putting on a happy face" actually works!) A renowned drama coach, Constantin Stanislavsky, had his actors carefully reproduce smiling and frowning as a way of creating these emotions on stage. Empathy (feeling someone else's emotional state) may happen because we unconsciously imitate another person's facial expressions. Whoever thought that making faces could reveal so much about the science of emotions!

"Making Mona Lisa Smile" by Nick D'Alto, ODYSSEY: Adventures in Space, November 2006. © by Ceres Publishing Company. Reproduced with permission.
All Cricket Media material is copyrighted by Ceres Publishing Company, d/b/a Cricket Media, and/or various authors and illustrators. Any commercial use or distribution of material without permission is strictly prohibited. Please visit <https://www.cricketmedia.com/info/licensing> for licensing and <https://www.cricketmedia.com> for subscriptions.

Prewriting/Planning

Use the blank Prewriting/Planning space below for notes, lists, webs, outlines, or anything else that might help you plan your writing. If you need additional space for planning, raise your hand and your teacher will give you a sheet of scratch paper. **You must write your final draft in the box on the right.**



Now check your writing using this Editing Checklist.

Editing Checklist

1. Check your capitalization and punctuation.
2. Spell all words correctly.
3. Check for sentence fragments or run-on sentences.
4. Keep verb tense consistent.
5. Make sure subject and verb agree.
6. Use words according to the rules of Standard English.
7. Remember to paragraph correctly.

Making Mona Lisa Smile

Prewriting/Planning

Checklist

Read the article "Making Mona Lisa Smile." Then answer the question.

Making Mona Lisa Smile

by Nick D'Alto



/ Alamy Stock Photo

In the article "Making Mona Lisa Smile," the author describes how a new technology called the Facial Action Coding System enables computers to identify human emotions. Using details from the article, write an essay arguing whether the use of this technology to read the emotional expressions of students in a classroom is valuable.

Be sure to include

- a claim for or against the value of using this technology to read students' emotional expressions
- evidence from the article to support the claim
- an introduction, a body, and a conclusion to your essay

Final Draft

10000